

Does Reciprocity Persuade?

Experimental Evidence from Doorstep Conversations

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Can conversational reciprocity enhance persuasion? While parties often reach out to a sizeable share of citizens through canvassing campaigns, doorstep conversations usually follow a topical script developed by the campaign. This leaves little room for a genuine exchange of perspectives. From a representation point of view, this is lamentable because such interactions forgo the potential to deepen mutual understanding between voters and representatives. From a persuasion perspective, conversational reciprocity might facilitate attitude change by increasing reception and acceptance of the persuasive message. Cooperating with a political party in Germany, this pre-registered field experiment tests whether a reciprocal approach to doorstep conversations — letting the voter talk about their key issues before introducing a topic important to the party — increases the propensity to vote for the party compared to a more traditional, guided conversation. Across ≈ 600 doorstep conversations, results show that reciprocal conversations increase the propensity to vote for the party by about 21% of a standard deviation ($p=0.09$). Effects on conversation tone and the importance voters assign to the party's topic are positive but all p-values cross the pre-registered threshold of 0.05. While imprecise, the overall findings suggest that increasing conversational reciprocity in canvassing campaigns can enhance persuasion. By leaving more room for genuine political exchange, parties might be able to better connect with citizens and improve the quality of democratic representation.

Table of contents

1	Notes for LESS	3
2	Introduction	4
3	Theoretical Background	4
3.1	The Active Ingredient of “Deep Canvassing”	4
3.2	Reciprocity in Doorstep Canvassing	8
3.2.1	Reception Reciprocity	8
3.2.2	Social Influence Reciprocity	9
4	Research Design	10
5	Data and Randomisation	12
5.1	Data Collection	12
5.2	Dependent Variables	12
5.3	Randomisation	13
6	Results	15
6.1	Main Effects	16
6.2	Heterogeneous Treatment Effects	16
6.3	Effects on Canvassers’ Evaluation	18
7	Robustness Checks	20
7.1	Exclude Guessed PTV	20
7.2	Lee Bounds	20
8	Discussion	22
9	References	23
10	Appendix	27
10.1	Balance and Placebo Tests	27
10.2	Canvasser Evaluations (H4–H7)	27
10.3	HTE Robustness Specifications (H8a–H8b)	27
10.4	Missing Data Sensitivity Analysis	31

1 Notes for LESS

Dear LESS participants,

This is an early draft of a pre-registered field experiment on conversational reciprocity in doorstep canvassing. I would be very grateful for your thoughts, pushback, and suggestions.

In particular, I would appreciate feedback on the following:

- **Structure and presentation of results:** The main hypothesis (H1: PTV) finds suggestive support, but all analyses are underpowered given the achieved sample size — yet the pre-registration commits me to reporting all specified tests. How would you advise structuring a paper that honestly presents these constraints while still making a clear where the contribution lies?
- **The reciprocity argument:** Does the theoretical framing resonate? Are there related literatures or specific works you think should be engaged with more directly?
- **Design improvements for a replication:** What would you change about the design — particularly with regard to the social desirability problem inherent in canvasser-reported PTV — if I were to run a larger follow-up study?
- **General framing:** I see two main frames for this paper but find it hard to commit to one. One is “looking for the active ingredient of deep canvassing”, the other starts from reciprocity as a general feature of political representation and persuasion and is somewhat more exploratory. Which one do you find more appropriate for this paper? Any other links that came to your mind when reading it?

Thank you so much for your time and help!

2 Introduction

This paper asks whether increasing conversational reciprocity in doorstep canvassing improves political persuasion. In particular, I study whether voters become more willing to support the partner organisation when canvassers first engage with voters’ own political views before priming a topic important to their party.

The main argument is that reciprocity is an active ingredient of *deep canvassing* interventions that involve high-quality non-judgemental listening. When canvassers listen attentively and signal that they take voters’ perspectives seriously, voters may feel a social obligation to reciprocate by being more open to the canvasser’s subsequent message. This perspective implies that not only message content, but also the interactional structure of the conversation, should shape persuasive success.

To test this claim, I conduct a pre-registered field experiment in collaboration with a political party in Germany. At the doorstep level, conversations are randomly assigned either to a reciprocal script (first eliciting and listening to voters’ broader views, then introducing a campaign topic) or to a less reciprocal script that starts directly with the campaign topic; both conditions include non-judgemental listening once the topic is discussed. Across ≈ 600 conversations, the treatment increases the propensity to vote for the party by 0.29 points on a five-point scale, corresponding to about 0.21 standard deviations ($p = 0.09$), providing support for the main hypothesis that reciprocity facilitates persuasion.

3 Theoretical Background

3.1 The Active Ingredient of “Deep Canvassing”

In the golden age of mass parties, direct face-to-face interactions between local party activists and the wider population were crucial for mobilising supporters. There are many things to dislike about the traditional party *machine*. Dense interactions and dependence on the machine often blurred or even crossed the line between programmatic representation and clientelism, and the deep entrenchment of party politics in local government is often credited with producing inefficient administration. The slow but steady decline of membership-based parties in the second half of the 20th century (Van Biezen, Mair, and Poguntke 2012; Scarrow 2014), which unfolded alongside a shift towards more centralised control over campaign strategies and

an increased reliance on advertising (Fourinaies 2021; Scarrow 2003), has deemphasized this traditionally dominant channel. The ground game was replaced by a much more diffuse and heavily mediated style of political communication. A national conversation driven by polls and aggregate opinion replaced the conversation on the street.

Direct face-to-face interactions between activists and voters saw a revival in the 1990s (and later in some European states), when researchers and parties started trialling and evaluating doorstep canvassing (Huckfeldt and Sprague 1987; Green, Gerber, and Nickerson 2003; Gerber and Green 2000). An early insight from these studies was that “the more personal the interaction between campaign and potential voter, the more it raises a person’s chances of voting” (Green and Gerber 2024, 10). With doorstep canvassing as the most personal strategy available, a large body of literature evolved to study why and when doorstep interactions are effective. The explored mechanisms include canvasser-voter-dyad properties like shared ethnicity or neighbourhood (Sinclair, McConnell, and Michelson 2013), message content (types of information and social pressure), and voter properties (turnout propensity, age, status). Yet, even though some degree of heterogeneity has been observed along a few of these dimensions, the broader insight was that “the message doesn’t matter much” (Green and Gerber 2024, 34). The key ingredient of a mobilisation message seems to simply be an explicit ask to turn out to vote. The only notable exception to this rule seems to be the activation of social norms: when voters are reminded of the civic duty to vote or that their adherence to this norm will be monitored, treatment effects are substantially larger (Gerber, Green, and Larimer 2008; Foos et al. 2026).

Concluding that the content of the interaction doesn’t matter much, academics worked out how many doorstep conversations translate into one additional vote (cf. Green and Gerber 2024, 172): if all that counts is the personal touch and some encouragement to vote, the main challenge is to deliver these in the shortest possible amount of time. In response, campaigns often optimised doorstep interactions to be as brief as possible, to exclude political talk as much as possible, and to follow narrow, pre-defined scripts (cf. SPD et al. 2013). At the same time, even though doorstep canvassing has become ever more widespread, its effectiveness has declined (Foos et al. 2026). While we can only speculate about the factors driving this development, it seems notable that recent research about doorstep *persuasion* (as opposed to mobilisation) has demonstrated strong effects of long, less scripted, high-quality conversations on voter attitudes (Kalla and Broockman 2020, 2023; Broockman and Kalla 2016). This is even more surprising given the longstanding finding that persuasion is far more difficult to

achieve than mobilisation (Kalla and Broockman 2018; Green and Gerber 2024, 184)¹

The treatments in Kalla and Broockman’s persuasion experiments are often characterised as *Deep Canvassing* strategies. Instead of a light-touch turnout reminder or even some not-so-light-touch social pressure, deep canvassing draws on an in-depth, non-judgemental conversation that focuses on the perspective of the voter. The canvasser takes a role akin to a facilitator in deliberative settings, actively listens to the voter’s views, asks open and clarifying questions, and seeks to create an atmosphere of mutual respect that is conducive to introspection. The interventions differ in the extent to which canvassers share their own views and the extent to which they structure the conversation around specific topics, and they usually represent large compound treatments. For example, in their 2020 field experiment on attitudes towards illegal immigration, the main treatment consisted of 8 steps: Contact, issue introduction, opinion elicitation, non-judgemental narrative exchange about immigration, non-judgemental narrative exchange about compassion, addressing voters’ concerns about immigration, making the case for compassion for undocumented immigrants, and finally a second elicitation of the voter’s views on immigration (Kalla and Broockman 2020, Appendix 1). Being exposed to this intervention vis-a-vis a placebo condition led to a treatment effect of around 8% of a standard deviation of the dependent variable. Notably, once the authors omitted the non-judgemental narrative exchange about immigration and compassion, the treatment effect disappeared, suggesting that the narrative exchange is the *active ingredient* through which the compound treatment works.

Still, even the *non-judgemental exchange of narratives* is a large compound: the canvasser conveys information and likely elicits their own preference, the voter thinks and talks about the issue and experiences non-judgemental listening. Naturally, the question arises which one of these components is fundamentally responsible for the persuasion. Social-psychologists have proposed an *introspection mechanism* according to which the non-judgemental *listening* – which forms part of any narrative exchange – reduces social anxiety in the speaker. This in turn reduces defensive processing: when the speaker doesn’t feel their attitudes are being attacked, they are more likely to be self-critical and engage in introspection which may in turn lead to more nuanced belief systems (Itzhakov, Kluger, and Castro 2017; Itzhakov et al. 2024). However, evidence for this mechanism is thus far limited to lab experiments that don’t test (political) persuasion effects.

Kalla and Broockman (2023) test three different strategies to identify the active ingredient of

¹Note that directly comparing the efficacy of these newer persuasion interventions to mobilisation treatments is not possible as the former haven’t yet been conducted in high salience general election settings.

the narrative exchange. They find that *perspective-getting* – being told by a canvasser about the experiences of an outgroup member – reduces exclusionary attitudes towards this outgroup, while other strategies that ask the voter to self-generate a story either about an outgroup member or about their own experiences of ostracism don’t cause attitude change. To further unpack this finding, they test whether perspective-getting affects a range of hypothesised mediators and find that it increases attributional thinking (attributing blame for outgroup’s challenges not to the outgroup), reactive empathy, and self-outgroup merging (feeling more similar to the outgroup).

In a recent lab-in-the-field experiment, Santoro et al. (2025) further randomise whether a persuasive appeal is delivered together or without high-quality non-judgemental listening and find that the persuasive message alone was similarly effective with and without listening. Interestingly, even though listening didn’t facilitate persuasion, it did lead to the speaker liking the listener more and to a reduction in speaker defensiveness.²

However, there are good reasons to question whether these findings generalise to the real world interactions we care about. Lab experiments with paid participants artificially secure engagement: unlike in a real-world conversation, participants are being paid for the interaction. Even if they dislike their interlocutor or don’t feel like sharing their perspectives, they will still do so to conform with the instructions. In real world settings, one of the biggest obstacles to any persuasive conversation is to simply *have* a conversation, to motivate the interlocutor to engage in the reflective exercise, or to listen to a lengthy narrative about an outgroup, political issue, or personal experience, let alone to share one.

Furthermore, the Santoro et al. (2025) experiment severs the link between the two interlocutors: in the entire interaction, it is clear and explicit that the persuader doesn’t act on their own volition. Participants are told that “the instructions” of the experiment dictate who gets to speak, who gets to listen, and which topic is being discussed. This likely creates a situation in which defensiveness does not present a major barrier to persuasion: if your interlocutor is not motivated to persuade you but only elaborates on their perspective because they are instructed to do so, there is less reason to defend one’s belief system. This doesn’t mean that defensiveness can’t be reduced through listening even in this setting (as the authors show), but it may already be so low that it doesn’t block processing. In fact, the authors’ summary

²This finding aligns nicely with contemporary research on persuasion outside the canvassing context, demonstrating that liking or disliking the source of an information doesn’t moderate the persuasive effect of the information (Coppock 2022; Broockman and Kalla 2025) While these findings point to information as the sole driver of persuasion, there are good reasons not to rush to this conclusion.

statistics support such an interpretation; even respondents who only heard a persuasive appeal didn't report feeling defensive (1.64 on a 5-point scale) (Santoro et al. 2025, Appendix Table S14).

3.2 Reciprocity in Doorstep Canvassing

While the mobilisation literature has studied the role of social norms extensively, this perspective hasn't received similar attention in the doorstep-persuasion literature. In this section, I argue that *reciprocity*, one of the most widespread and pervasive social norms in human societies, is the active ingredient that renders some doorstep interactions more persuasive than others.

Reciprocity is a generalised social norm which demands that voluntarily given favours (or disfavours) are returned, even if such behaviour is costly (cf. Gouldner 1960; Fehr and Gächter 2000). The norm of reciprocity is the reason we feel an obligation to invite friends to our birthday parties if they invited us to theirs. The norm is pervasive across diverse cultures and operates at all levels of politics: among elites that reciprocate favours as well as at the grassroots where activists make support for representatives conditional on political actions or (in the clientelist setting) material benefits (Cialdini 2013).

I theorise that deep-canvassing interactions (like non-judgemental listening) – in which volunteers take time and make an effort to listen to voters' views – are effective because they initiate a reciprocal interaction. Reciprocity operates through two channels: reception reciprocity and social influence reciprocity.

3.2.1 Reception Reciprocity

According to Zaller (1992), for a message to be persuasive, it needs to be 1) received and 2) accepted. A persuasive message will thus only enter an individual's considerations if the individual was *exposed*, managed (and was willing to) *receive* the message, and was unable to refute its content (*accept*). I argue that reciprocity can operate at both key stages. In deep-canvassing interventions, the persuasive message is preceded by an episode of attentive, emphatic listening. The voter experiences a canvasser who engages with their worldview and signals respect and interest. In short, the canvasser ostensibly *receives* the voter's message.

A reciprocity mechanism would suggest that the speaker notices this – perhaps surprising – attention and feels a social obligation to reciprocate it by being more receptive to the canvasser’s message. Listening activates reciprocity norms and makes the voter more likely to receive the subsequent persuasive message. Note that conditioning on reception (as most lab experiments do) makes this mechanism impossible to detect. If you ever felt like you at least had to listen to a friendly fundraiser’s pitch after they helped you pack your groceries into the car, you have experienced this mechanism in action.

3.2.2 Social Influence Reciprocity

The second mechanism operates at the acceptance stage and is both more demanding and more interesting from a democratic representation perspective. Analogously to the reception mechanism, the social influence reciprocity mechanism suggests that when we feel that someone is influenced by our views, we feel a social obligation to reciprocate this influence by being more open to their views as well. The mechanism is more demanding because it requires a setting in which ego perceives whether alter is influenced by their views. While perceiving alter as receptive might cause a reduction in defensiveness and an increase in liking, the persuasive effect would not travel through these channels. Instead, sensing that alter is influenced by ego’s views would cause ego to feel a social obligation to reciprocate this influence by being more open to alter’s views.

Mahmoodi, Bahrami, and Mehring (2018) test this mechanism in a lab experiment in which participant A is asked to memorise the location of a dot on a screen. A makes their own guess and is then shown what they believe to be another participant’s (B) guess. B then has the opportunity to revise their guess after seeing A’s guess. Finally, A gets to update their initial guess. The authors find that A’s revision is more influenced by B’s guess when B was influenced by A’s initial guess. Furthermore, these effects disappear once A is informed that B is a computer and not a human. In other words, participants were more susceptible to social influence when they felt that their own views were also being considered and influenced the other person. The fact that this effect only emerged when participants believed they were interacting with a human suggests that reciprocity, not some other mechanism, is at play.

Social influence reciprocity has important implications for democratic representation. Political theorists have long argued that democratic representation should be optimised to facilitate mutual understanding between representatives and the represented (Mansbridge 2003; Disch

2011). This criterion, often referred to as “reflexivity”, is grounded in the idea that representatives can learn genuinely new and useful information from their constituents, while voters’ preferences are often far from stable, fully formed, and well-considered. Because voters are not always aware of the intricacies, interconnectedness, and trade-offs of political issues, they might benefit from deliberation with their more politically knowledgeable representatives (cf. Mair 2009).

In the remainder of this paper, I describe the research design and findings of a field experiment that tests key hypotheses derived from the theory above.

4 Research Design

This experiment tests whether increasing the degree of conversational reciprocity in doorstep canvassing conversations facilitates persuasion, engagement with the topic introduced by the canvasser, and the overall tone of the interaction. In the treatment group, canvassers follow a reciprocal listening script with two phases: after a short introduction, the canvasser asks the voter to elaborate their political views and listens actively, using non-verbal and verbal cues to signal their engagement with the voter’s perspective. This is phase 1. In phase 2, the canvasser introduces one of three randomly assigned topics (fly tipping and local environment, education and schools, or transport and mobility) and asks the voter about their views on this topic. In the control group, canvassers jump right to the second phase: after a short introduction, they introduce the randomly assigned topic and ask the voter’s views on it, without first eliciting and engaging with the voter’s broader political views. Crucially, both treatment and control scripts include active listening.

Heterogeneous treatment effects are explored for demographic similarities between canvassers and voters. Specifically, I study whether the above-mentioned effects are stronger when canvasser and voter are of the same age group or gender.

In addition to testing the effects of reciprocal listening on voters, the experiment analyses whether canvassers’ exposure to reciprocal listening affects their evaluation of the canvassing experience, namely their enjoyment of the activity, their self-assessed ability to represent, persuade, and understand voters in the canvassed neighbourhood. Future analyses will also explore the effects of reciprocal listening dosage on canvasser retention.

The pre-registered hypotheses are summarised in Table 1. The pre-registration is available [here](#).

Table 1: Pre-registered Hypotheses

Hypothesis	Description
H1	Voters exposed to reciprocal listening will assign higher importance ratings to the subsequently introduced party topic than voters in the unidirectional condition.
H2	Voters exposed to reciprocal listening will report a higher propensity to vote for the partner organisation than voters in the unidirectional condition.
H3	The reported tone of reciprocal conversations will be better than that of unidirectional ones.
H4	Activists with higher exposure to reciprocal listening evaluate their experiences more positively.
H5	Activists with higher exposure to reciprocal listening report having a better understanding of their interlocutors.
H6	Activists with higher exposure to reciprocal listening report greater confidence in their ability to represent constituents in the canvassed neighbourhood.
H7	Activists with higher exposure to reciprocal listening report greater confidence in their persuasive capabilities.
H8	Activists evaluate conversations more positively when their interlocutor is similar to them in demographics.
H8a	Activists evaluate conversations more positively when their interlocutor is of the same gender.
H8b	Activists evaluate conversations more positively when their interlocutor is of the same age group.
H8c	Activists evaluate conversations more positively when their interlocutor is of the same age group and gender.
H9	Canvassers will be more likely to return to the next canvassing session if they were in the high-reciprocity condition last time.
H10	The time between two sessions a canvasser participates in will be shorter if the canvasser was in the high-reciprocity condition in the first session.

5 Data and Randomisation

5.1 Data Collection

Data collection takes place through three dedicated surveys. 1) A baseline survey of activists that each canvasser fills out before their first canvassing session, 2) a web-app like qualtrics survey used to collect data about each conversation at the doorstep as well as post-session evaluations from the canvassers, and 3) a post-campaign survey of activists conducted at the end of the campaign. This final survey mirrors the baseline survey and additionally collects feedback on organisational aspects of the campaign. The endline survey has not yet been conducted.

5.2 Dependent Variables

Table 2 summarises the dependent variables collected and analysed in this report. Doorstep-level variables are collected at the end of each conversation, while session-level variables are collected at the end of each canvassing session. PTV and Topic Importance are measured through the canvasser, who asks the voter the respective questions and records their answers. If the voter refuses to answer or the conversation ends before the question is asked, the canvassers are instructed to make a best guess about PTV (but not topic importance) and to indicate in a separate question that the answer is a guess.

Table 2: Measured Variables

Variable Label	Question Text	Response Scale	Level
Door Opened	Did someone open the door?	Binary: Yes/No	Door
Conversation Occurred	Did a conversation take place?	Binary: Yes/No	Door
Conversation tone	How was the tone?	5-point scale: Very good to Very bad	Door

Variable Label	Question Text	Response Scale	Level
Topic Importance	How important was the topic [random_topic] to the person?	5-point scale: Very important to Very unimportant + Don't remember	Door
PTV (Partner Org.)	How likely is it that the person will support the [Partner Organisation] with their vote in the [next election]?	5-point scale: Very likely to Very unlikely + Don't remember + Not eligible to vote	Door
Age Estimate	How old do you estimate the person to be?	6 categories	Door
Gender	What gender was the person?	4 categories: Female/ Male/ Diverse/ Don't know	
Overall Satisfaction	How did you like the door-to-door canvassing today?	5-point scale: Very good to Very bad	Session
Understanding Positions	How well could you understand your interlocutors' positions?	5-point scale: Very good to Very bad	Session
Representation Ability	How well could you personally represent the people in the neighbourhood?	5-point scale: Very good to Very bad	Session
Persuasiveness	How well could you persuade people in today's conversations?	4-point scale: Very good to Very bad	Session

5.3 Randomisation

The experiment uses two levels of randomisation. Figure 1 illustrates the design.

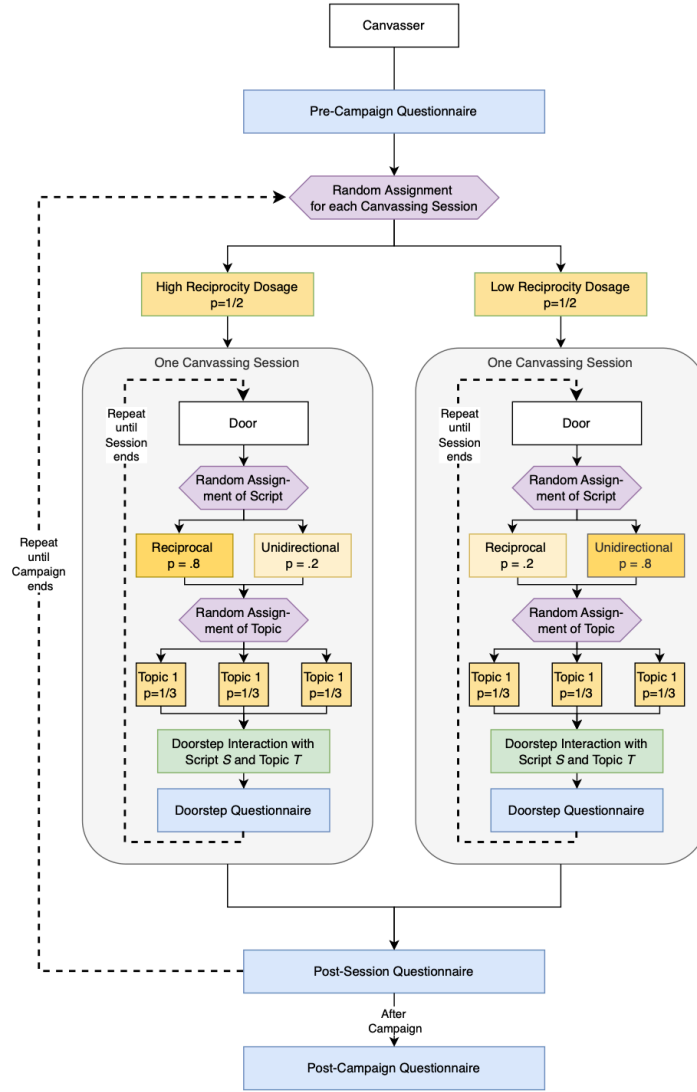


Figure 1: Experimental Design

At the **session level**, each canvasser is randomly assigned to a *high-reciprocity* or *low-reciprocity* condition at the start of each session (50:50). This session-level assignment determines the mix of scripts used over the course of the session: roughly 80% of doors receive the reciprocal script in the high condition and roughly 20% in the low condition.

At the **door level**, the Qualtrics tool independently randomises a script (reciprocal or unidirectional) and a topic (fly tipping and local environment, education and schools, or transport and mobility, each with equal probability) before the canvasser approaches each door.³ The door-level treatment indicator is the primary quantity used in the main analysis: each door is assigned independently of previous doors and regardless of the canvasser’s session-level condition.

If a canvasser moves to a different neighbourhood or canvasses on a different day, a new session assignment is drawn. Multiple canvassers working in the same neighbourhood simultaneously receive independent session assignments.

6 Results

To estimate the average treatment effect of the reciprocal listening treatment, I fit linear regression models according to the specifications in Equation 1:

$$Y_{ijk} = \tau_{\text{ATE}} D_{ijk} + \gamma_{jk} + \varepsilon_{ijk} \quad (1)$$

Where Y_{ijk} is the outcome of interest for voter i canvassed by canvasser j in session k , D_{ijk} is a binary indicator for treatment assignment, γ_{jk} are canvasser-session fixed effects, and ε_{ijk} is an error term. I control for canvasser-session fixed effects to account for between canvasser-session differences in assignment probabilities (Humphreys 2025, 4; Gerber and Green 2012, 119) and to improve the precision of the estimates.

³The tool pre-generates up to 150 door assignments per session (the maximum it can record). If this limit is reached, the tool submits the data and starts a new session.

6.1 Main Effects

The main results for hypotheses H1 to H3 are summarised in Table 3. Results indicate that the treatment had a positive and significant effect on the propensity to vote for the partner organisation (PTV). The effect is 0.29 on a 1-5 scale, which corresponds to an increase of ≈ 0.21 standard deviations of the dependent variable in the control group. The effects on conversation tone and topic importance are positive but not statistically significant at conventional levels. It should be noted that the tone of most conversations was rated extremely positively (average of 4.26 on a 1-5 scale with 5 being “very good”), which means that ceiling effects make it difficult to detect differences between treatment and control conversations. Similarly, the average topic importance rating was 3.77. Propensity to vote for the partner organisation had the lowest average rating of all three outcomes (3.38).

Given that PTV and topic importance were measured through the canvasser at the end of the conversation, it cannot be ruled out that systematic non-response patterns bias the results. To explore this, Table 14 presents bounds on the treatment effect for PTV.

6.2 Heterogeneous Treatment Effects

Next, I explore heterogeneous treatment effects by demographic similarity between canvassers and voters. Because canvassers typically worked in pairs and the data-entry tool initially recorded only one team member’s identity, the pre-registered similarity measure is ambiguous: it reflects the characteristics of whoever entered the data, not necessarily the canvasser who led each conversation. Teams were deliberately composed to be demographically dissimilar — typically pairing a younger with an older canvasser, and a male with a female canvasser — so attributing each doorstep to the data-entry canvasser introduces systematic measurement error in opposite-gender and mixed-age dyads. The analysis below therefore restricts to *homogeneous canvasser teams*: sessions in which both team members share the demographic characteristic of interest, so the similarity measure is unambiguous regardless of which team member led the conversation. Results from first-canvasser specifications (ignoring the second canvasser and only considering the data-entering canvasser) is reported as a robustness check in Section 10.3.

Voter age is measured through the canvasser’s estimation in six pre-defined categories (18–24, 25–34, 35–44, 45–54, 55–64, 65+); canvasser age was measured at baseline. Age similarity is defined as canvasser and voter falling in the same category. One potential concern is reverse

Table 3: Effects of Reciprocity on Conversation Outcomes

	PTV	Conversation Tone	Topic Importance
D	0.285+	0.091	0.149
	p=0.094	p=0.270	p=0.275
	[−0.043, 0.623]	[−0.070, 0.252]	[−0.113, 0.418]
R2	0.22	0.19	0.21
P-value type	two-tailed	two-tailed	two-tailed
CI Level	0.95	0.95	0.95
Y0_sd	1.329	0.881	1.214
Y1_sd	1.433	0.827	1.208
N	465	679	544
N_D0	262	365	314
N_D1	203	314	230
Canvasser-Session Fixed Effects	Yes	Yes	Yes

+ p < 0.1, * p < 0.05, ** p < 0.01, *** p < 0.001

Confidence intervals and p-values based on randomisation inference with 10000 permutations.

causality: if canvassers perceive more receptive voters as more similar, the interaction term would be biased upward. Reassuringly however, balance tests in Table 8 show no systematic differences in reported voter age or gender as a function of treatment assignment.

A structural feature of the data shapes the gender analysis in particular. All canvassing teams were either mixed-gender (one male, one female) or all-female; there were no all-male teams. The homogeneous-team restriction therefore retains only observations from all-female teams. Within this subset, `team_gender_match` equals one when the voter is also female. The interaction term in Table 4 should be interpreted as the differential treatment effect when a female team canvasses a female voter relative to a male voter — not a symmetric same-gender versus opposite-gender comparison.

Results in Table 5 and Table 4 show no significant heterogeneity in treatment effects by age similarity or gender configuration.

Pre-registered H8c — simultaneous similarity in both age and gender — cannot be tested with the available data. Within the homogeneous-team subset, the joint both-match condition (same age category and female voter for an all-female team) reduces to too few observations

with no usable treatment variation. These cell sizes are insufficient for a reliable RI F-test of the three-way interaction, so H8c is omitted from the analysis.

6.3 Effects on Canvassers' Evaluation

Results for H4–H7 are reported in Table 9. The unit of analysis is the canvasser-session ($N = 68$). The pre-analysis plan anticipated that these tests would be underpowered even at the target sample of 200 sessions; with the current sample, all four outcomes — satisfaction, representativeness, understanding, and persuasiveness — yield null results. No strong conclusions can be drawn from these estimates.

Pre-registered H9 and H10 concern canvasser retention: whether high-reciprocity dosage predicts returning to the next session (H9) and a shorter gap between sessions (H10). H10 cannot be estimated because fewer than three canvassers participated in more than two sessions, leaving no usable variation in the outcome. H9 can be estimated with $N = 51$ canvassers, but the standard error is more than twenty times the point estimate, making the result wholly uninformative. Given that the sample of canvasser-sessions is less than half the pre-registered target of 200, both results are unsurprising.

Table 4: Heterogeneous Treatment Effects by Canvasser Team Gender

	PTV	Conversation Tone	Topic Importance
D x Same Gender (team)	0.741	0.523	0.189
Same Gender (team)	−0.607	−0.311	−0.539
D	0.513	−0.100	0.172
F-statistic	1.042	1.974	0.124
RI: F-stat p-value	0.198	0.155	0.661
RI: Sims	10000	10000	10000
N	68	145	82
Session Date FE	Yes	Yes	Yes
Neighbourhood FE	Yes	Yes	Yes

Coefficients from the unrestricted model; 95% confidence intervals in brackets.

Table 5: Heterogeneous Treatment Effects by Canvasser Team Age Group

	PTV	Conversation Tone	Topic Importance
D x Same Age Group (team)	−1.789	−0.059	0.203
Same Age Group (team)	1.255	0.198	0.089
D	0.389	0.084	0.836
F-statistic	4.602	0.016	0.122
RI: F-stat p-value	0.026	0.935	0.888
RI: Sims	10000	10000	10000
N	50	96	61
Session Date FE	Yes	Yes	Yes
Neighbourhood FE	Yes	Yes	Yes

Coefficients from the unrestricted model; 95% confidence intervals in brackets.

7 Robustness Checks

7.1 Exclude Guessed PTV

If a voter refused to answer the PTV question or the conversation ended before the question was asked, canvassers were instructed to make a best guess about PTV and indicate in a separate question that the answer is a guess. To explore whether these guessed values bias the results, I re-estimate the main treatment effect excluding all observations where PTV was guessed. The results are summarised in Table 6 and indicate that the treatment effect barely changes. However, the p-value increases slightly to 0.096.

7.2 Lee Bounds

The sensitivity analysis above makes no assumption about *why* PTV is missing. A complementary approach due to Lee (2009) imposes only the weak assumption of *monotone selection*: the treatment shifts the probability of observing PTV in one direction only. Under this assumption, it is possible to derive sharp bounds on the treatment effect for “always-observed” voters — those who would report their PTV regardless of which script was delivered.

Within the analysis sample of open-door conversations, 64% of treated voters provided a PTV response, compared to 71.8% of control voters. The gap indicates that the control arm has excess selection of approximately 10.8 percentage points. Under monotone selection, this excess is attributed to “complier” voters who would only report PTV when assigned to control — voters who might, for instance, be unwilling to engage with a canvasser using the reciprocal script but willing to engage with the standard one. To bound the treatment effect for always-observed voters, the 10.8% of control observations most likely to be compliers are trimmed in turn from the top and bottom of the control outcome distribution (within block $\times$ session-day strata), yielding a lower and an upper bound.

Table 7 reports the estimates. Both bounds are positive and the lower bound remains marginally significant, indicating that the positive effect on PTV holds even for the subpopulation of voters who would engage with a canvasser irrespective of the script delivered. As a complement, Section 10.4 reports arm-blind sensitivity checks: for each category of missing PTV values, missing responses are replaced with the worst-case value (1) or the best-case value (5), and the main model is re-estimated (Coppock 2019). Unlike the Lee bounds, these scenarios impose no assumption about the direction of selection; they address whether the

Table 6: Robustness Check: Excluding Guessed PTV Values

	PTV
D	0.296+
	p=0.096
	$[-0.047, 0.645]$
R2	0.22
P-value type	two-tailed
CI Level	0.95
Y0_sd	1.34
Y1_sd	1.437
N	446
N_D0	252
N_D1	194
+ p <0.1, * p <0.05, ** p <0.01, *** p <0.001	

result survives extreme assumptions about what missing voters would have reported. The treatment effect for PTV remains positive across all scenarios.

Table 7: Lee (2009) Bounds on the Treatment Effect on PTV

	Lower bound	Upper bound
D (Treatment)	0.123	0.491**
	p=0.488	p=0.004
	[−0.228, 0.474]	[0.159, 0.823]
Num.Obs.	426	426
R2	0.225	0.261
FE: ‘Canvasser-Session‘	X	X
Canvasser-Session FE	Yes	Yes

+ p < 0.1, * p < 0.05, ** p < 0.01, *** p < 0.001

Lower bound: trims bottom 10.8% of control outcomes.

Upper bound: trims top 10.8%.

8 Discussion

This paper tested whether inserting a reciprocal listening phase — asking voters to share their own views before introducing the party’s topic — increases persuasive success relative to a standard unidirectional canvassing script. Both conditions included high quality listening. The results are inconclusive: while the treatment seems to have increased PTV, the effect is not significant at the pre-registered threshold of 0.05 and imprecisely estimated (95% CI: −0.043, 0.623). Yet, given the experiment didn’t reach the envisaged sample size, the point estimate is far from zero, consistently positive across bounding analyses, and significant at the 0.1 level, dismissing the results would be too radical. Instead, the findings should be as interpreted as suggestive evidence that reciprocity may be a key mechanism of persuasion in doorstep canvassing, and as motivation for a larger-scale replication to draw more definitive conclusions.

This paper doesn’t propose reciprocity as a new means of persuasion but advances the argument that it is a thus far overlooked mechanism that explains the persuasive power of a range of interactions. Many of the interventions trialed in the persuasion and deep-canvassing literature involve some form of reciprocity, even though it’s existence is often neither theorised nor tested. As discussed above, reciprocity could explain why non-judgemental listening facilitates persuasion in field settings but not in lab experiments, where it is performed in exchange for payment by the researcher, not as a voluntary favour to the interlocutor.

One obvious objection against the experimental protocol is that PTV is recorded by the canvassers themselves. Voters might systematically overstate their PTV to a sympathetic canvasser, thus inducing social desirability bias. While this is perfectly possible, it would only bias the ATE if the reciprocal listening condition induced *more* social desirability bias than the unidirectional condition. If voters find it unpleasant to tell canvassers that they’d never vote for their party, that should induce the same degree of bias in both conditions, leaving the ATE unbiased. In contrast, it could well be that the reciprocity ingredient induces additional social desirability bias. If we define social desirability bias canonically as a preference falsification strategy to avoid social sanctions and discomfort, this would still imply that the bond formed through reciprocal listening is stronger, increasing the perceived cost of disappointing the canvassers expectation.

Finally, this paper outlined a novel experimental protocol that has two main practical advantages that could facilitate field access and research efficiency. Firstly, by cross-randomising the topics pitched by canvassers at the doorstep, the experiment also gained insights into the effectiveness of three topics the cooperation partner could see itself campaigning on. Such campaign message evaluation is something many (local) political campaigns can’t afford to conduct and offering it to cooperation partners could produce win-win situations that facilitate field access for experimentalists. Secondly, while underpowered in this application, the session-level randomisation of treatment dosage alongside post-session canvasser evaluations allows researchers to study effects on voters *and* canvassers. Given the relevance of first-stage mobilisation (Neuenschwander and Foos 2021) for the scalability and sustainability of volunteer driven campaigns, understanding how different canvassing scripts affect canvasser motivation and retention is crucial for both researchers and practitioners.

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10 Appendix

10.1 Balance and Placebo Tests

Table 8 reports six balance and placebo tests. The first two columns check whether treatment assignment predicts door opening probability and conversation occurrence — outcomes that are determined before the script is delivered and should therefore be unaffected by treatment. The remaining columns test page duration (a proxy for how quickly canvassers completed the survey), conversations per canvasser, and the two demographic variables used in the HTE analysis (voter age group and voter gender). None of the estimates are statistically significant, providing no evidence of randomisation failures or systematic measurement differences between treatment and control.

10.2 Canvasser Evaluations (H4–H7)

Hypotheses H4–H7 ask whether delivering the reciprocal script affects how canvassers themselves evaluate the session: specifically, whether they found the conversation satisfying, whether they felt they represented the party well, whether they felt they understood the voter’s views, and whether they felt persuasive. These are canvasser-session-level outcomes, and the unit of analysis is accordingly the canvasser-session ($N = 68$), roughly half the pre-registered target. The pre-analysis plan acknowledged that these tests would likely be underpowered even at the target of 200 sessions. All four outcomes yield null results, and no strong conclusions can be drawn; the estimates should be read as consistent with either a genuine absence of canvasser effects or insufficient precision to detect them.

10.3 HTE Robustness Specifications (H8a–H8b)

The following tables report two alternative specifications as robustness checks for the homogeneous-team results in the main text. Table 10 and Table 11 use a first-canvasser specification, attributing each doorstep to the team member who entered the data. Because teams were typically mixed-gender and mixed-age, this specification introduces measurement error in opposite-gender and cross-age dyads — but it is reported here as the pre-registered default. Table 12 and Table 13 restrict to the final canvassing session, in which the lead canvasser at each door was recorded directly. Neither alternative changes the null finding.

Table 8: Balance and Placebo Tests

	Door Opened	Conversation Occurred	Page Duration (s)	Convos per Canvasser	Voter Age Group	Voter Gender
D (Treatment)	0.029	0.059	8.143	-0.461	-0.102	-0.010
	p=0.210	p=0.174	p=0.334	p=0.784	p=0.408	p=0.822
	[-0.016, 0.074]	[-0.027, 0.145]	[-8.538, 24.825]	[-3.831, 2.910]	[-0.343, 0.139]	[-0.096, 0.077]
Num.Obs.	2846	829	682	83	956	933
R2	0.054	0.158	0.456	0.632	0.138	0.089

+ p < 0.1, * p < 0.05, ** p < 0.01, *** p < 0.001

FE: respondent ID for voter-level models (1-3, 5-6); canvasser for model 4 (conversations per canvasser).

Table 9: Effects of High Reciprocity Dosage on Canvassers' Evaluation

	Like	Represent	Understand	Persuade
High Reciprocity Dosage	0.013	-0.155	0.060	0.089
	p=0.942	p=0.465	p=0.692	p=0.531
	[-0.344, 0.365]	[-0.562, 0.259]	[-0.241, 0.353]	[-0.185, 0.363]
R2	0.25	0.36	0.46	0.27
P-value type	two-tailed	two-tailed	two-tailed	two-tailed
CI Level	0.95	0.95	0.95	0.95
Y0_sd	0.746	0.902	0.703	0.608
Y1_sd	0.74	0.693	0.716	0.466
N	78	68	75	73
N_D0	41	37	41	41
N_D1	37	31	34	32
Session Date FE	Yes	Yes	Yes	Yes
Neighbourhood FE	Yes	Yes	Yes	Yes

+ p < 0.1, * p < 0.05, ** p < 0.01, *** p < 0.001

Confidence intervals and p-values based on randomisation inference with 10000 permutations.

Table 10: HTE by First-Recorded Canvasser Age Group (Pre-Registered Specification)

	PTV	Conversation Tone	Topic Importance
D x Same Age Group (first canvasser)	-0.097	0.253	0.236
Same Age Group (first canvasser)	0.398	0.085	-0.008
D	0.271	-0.039	-0.048
F-statistic	0.101	2.561	0.898
RI: F-stat p-value	0.877	0.477	0.515
RI: Sims	10000	10000	10000
N	454	927	527
Session Date FE	Yes	Yes	Yes
Neighbourhood FE	Yes	Yes	Yes

Coefficients from the unrestricted model; 95% confidence intervals in brackets.

Table 11: HTE by First-Recorded Canvasser Gender (Pre-Registered Specification)

	PTV	Conversation Tone	Topic Importance
D x Same Gender (first canvasser)	0.150	0.188	−0.160
Same Gender (first canvasser)	0.067	0.007	−0.017
D	0.176	−0.048	0.140
F-statistic	0.31	1.917	0.539
RI: F-stat p-value	0.564	0.175	0.697
RI: Sims	10000	10000	10000
N	440	904	512
Session Date FE	Yes	Yes	Yes
Neighbourhood FE	Yes	Yes	Yes

Coefficients from the unrestricted model; 95% confidence intervals in brackets.

Table 12: HTE by Lead Canvasser Age Group (Final Session)

	PTV	Conversation Tone	Topic Importance
D	−0.272	−0.233	−0.041
F-statistic	0.15	1.881	1.744
RI: Sims	10000	10000	10000
N	29	46	40
Session Date FE	Yes	Yes	Yes
Neighbourhood FE	Yes	Yes	Yes

Coefficients from the unrestricted model; 95% confidence intervals in brackets.

Table 13: HTE by Lead Canvasser Gender (Final Session)

	PTV	Conversation Tone	Topic Importance
D	−0.106	−0.061	0.545
F-statistic	0.02	0.014	0.284
RI: Sims	10000	10000	10000
N	29	46	40
RI: F-stat p-value		0.925	0.55
Session Date FE	Yes	Yes	Yes
Neighbourhood FE	Yes	Yes	Yes

Coefficients from the unrestricted model; 95% confidence intervals in brackets.

10.4 Missing Data Sensitivity Analysis

PTV responses are missing for four distinct reasons: no conversation took place (door not opened or voter refused to talk — canvassers were not shown the PTV question in this case), canvassers indicated they “don’t remember” the voter’s PTV, the voter reported being ineligible to vote, or the PTV question was simply not answered (true NA). Figure 2 shows the distribution of these categories across the sample.

Since conversation tone was recorded by the canvassers, it can be used to characterise the conversations with missing PTV. Conversations where the PTV was not recalled, where no reason for the missing value was indicated, and where no conversation took place had below-average tone; conversations with ineligible voters had above-average tone (Figure 3).

Table 14 presents the sensitivity analysis. The first row imputes missing PTV values using an ordered logit model regressing PTV on conversation tone. The remaining rows apply arm-blind worst- and best-case imputations for each missing-data category: the lower bound (NA = 1) asks whether the effect survives if all missing values are assigned the most negative possible outcome; the upper bound (NA = 5) asks the symmetric question. Because the imputations are arm-blind, the two columns are not formal Horowitz-Manski identification bounds — they can cross when a category is unevenly distributed across treatment arms. They should be read as sensitivity checks: does the direction and approximate magnitude of the treatment effect survive the most extreme assumptions? Across all rows, the answer is yes.

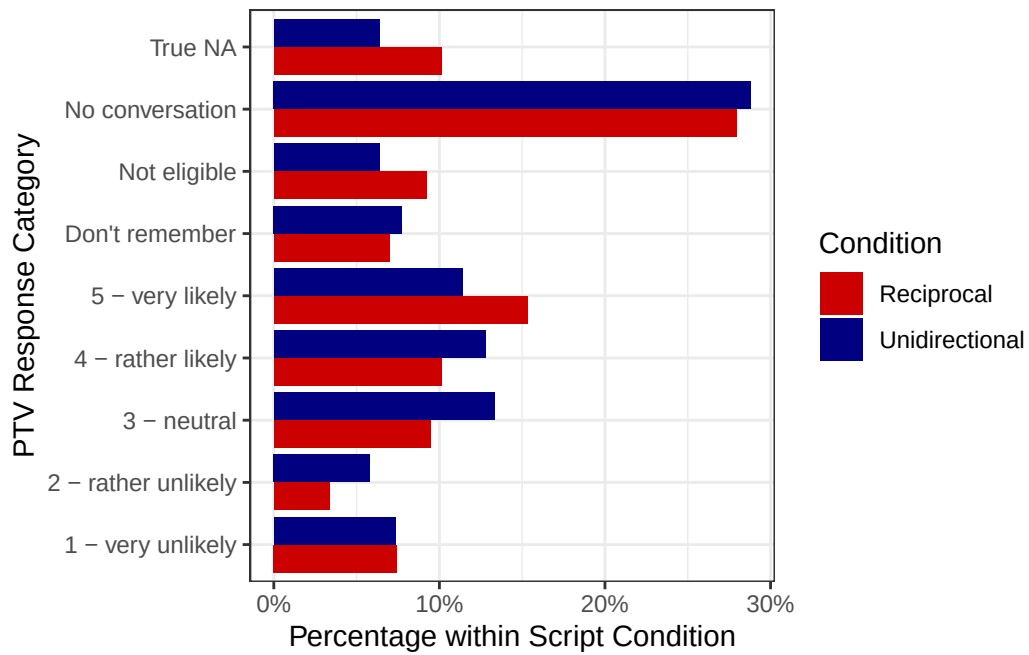


Figure 2: Missingness Pattern in PTV Responses

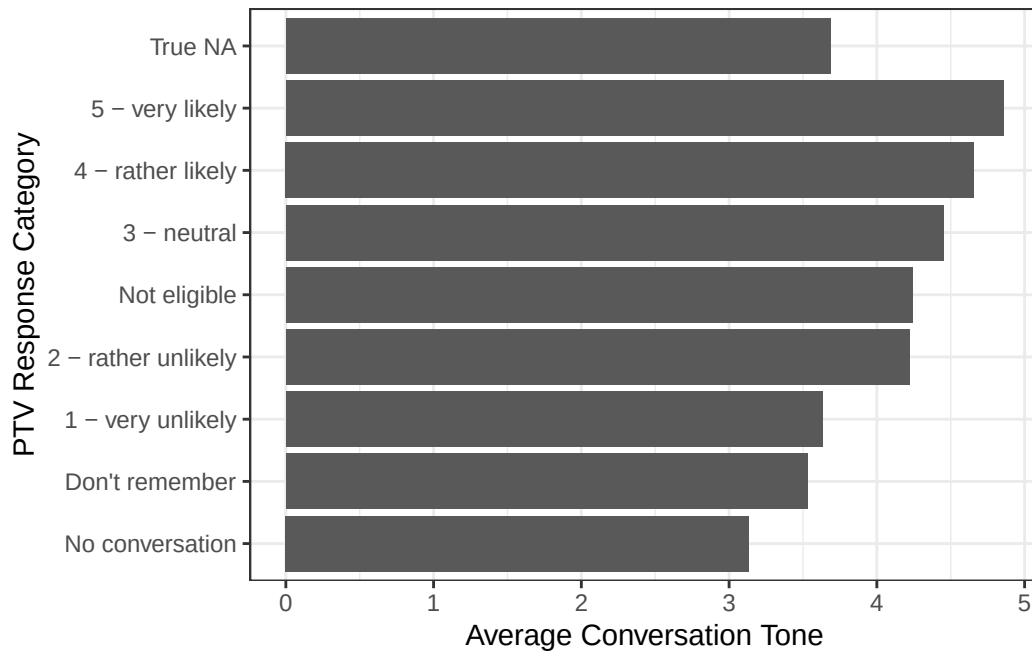


Figure 3: Average Conversation Tone by PTV Response Type

Table 14: Sensitivity Analysis: Treatment Effect Under Extreme Missing-Data Assumptions

	Mood imputation	NA recoded to 1	NA recoded to 5
Mood imputation	0.325*** (0.115)	—	—
Don't remember	—	0.335** (0.163)	0.186 (0.151)
Not eligible	—	0.072 (0.165)	0.433*** (0.153)
Any missing	—	0.012 (0.125)	0.259** (0.110)
True NA	—	0.332** (0.164)	0.203 (0.161)
No conversation	—	0.014 (0.145)	0.299** (0.124)
Canvasser-Session FE	Yes	Yes	Yes
N	961	961	961

^a * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. Asymptotic standard errors clustered at the canvasser-session level in parentheses.

^b Mood imputation: missing PTV values predicted via ordered logit on conversation tone; no directional worst/best-case counterpart.

^c Lower bounds recode missing PTV as 1 (worst case); upper bounds recode as 5 (best case).